

G01 - Skin Lesion Classification via Transfer Learning and Ensembling on the ISIC 2019 Dataset

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Abstract—The accurate classification of skin lesions is critical for the early detection of melanoma and other malignancies. Using the ISIC 2019 dataset, we evaluated three pretrained convolutional architectures — EfficientNet-B0, ResNet-50, and MobileNetV3-Small — adapted via transfer learning for the classification of eight-class lesion. We apply systematic ablation and hyperparameter optimization (HPO) to improve recall on minority classes. An ensembling study compares all variants on a panel of metrics including balanced accuracy (BACC), macro F1, and class-specific AUC to identify the most clinically effective approach for automated skin lesion triage. Code is available at <https://github.com/virginia-vrb9e/isic2019-skin-lesion-classification>.

Index Terms—Skin Lesion Classification, CNN, ResNet50, Transfer Learning, Ensembling, ISIC 2019

I. INTRODUCTION

According to the American Cancer Society, more than 5,000,000 cases of skin cancer are diagnosed annually in the United States [1], [2]. Although the vast majority are associated with low mortality rates, new cases of melanoma, the deadliest form of skin cancer, have increased dramatically in recent decades, with an estimated 200,000 new cases expected in 2026 [3]. However, early detection of melanoma is associated with a 5-year survival rate of more than 95%, creating a dire need for better and more efficient diagnostic tools.

One organization in particular, the International Skin Imaging Collaboration (ISIC), takes a global lead in establishing industry imaging standards while supporting active research. Inspired by the ISIC benchmark challenge series, we chose the ISIC 2019 dataset with published test ground truth values so we could perform a comprehensive analysis of our models. Throughout, our motivation and research question examines how imbalance strategies and ensembles of fine-tuned CNN models can improve minority-class recall in skin lesion classification in a commodity GPU environment.

II. DATASET

The ISIC 2019 aggregate dataset contains 33,569 images and associated metadata compiled from three sources for the 2019 benchmark challenge. The data is pre-split into a training set of 25,331 and a test set of 8,238 JPG images. Our working dataset has eight classes and is severely imbalanced - melanoma (MEL) 17.9%, melanocytic nevus (NV) 50.8%,

basal cell carcinoma (BCC) 13.1%, actinic keratosis (AK) 3.4%, benign keratosis (BKL) 10.4%, dermatofibroma (DF) 0.9%, vascular lesion (VASC) 1.0%, and squamous cell carcinoma (SCC) 2.5% - where the NV class comprises over 50% of the training data while DF and VASC together comprise less than 2%. We address this severe imbalance through class-weighted cross-entropy with inverse frequency weighting and report balance accuracy (BACC) and per-class AUC.

The ISIC 2019 dataset also includes a 9th class, "UNK", of 2,047 images, which we will not include in our analysis. It represents skin lesions that are not one of the eight identified classes and would be a necessary component of more complex analysis or if images were added to our working dataset in order to either augment minority classes or improve the bias towards Fitzpatrick 1 - 3 skin tones. It should be noted that the majority of the ISIC 2019 data is comprised of Fitzpatrick 1 - 3 skin tones, which is a troubling source of bias in models as it makes them less robust towards variances of underlying skin tone [18].

III. LITERATURE REVIEW

Our literature review began with a 2014 paper [8] that used a modified Falcon Finch deep convolutional neural network (CNN) to improve accuracy and computational efficiency using a dataset of 3,297 skin lesion images. Feature extraction was performed using ResNet-101 deep features and fed into a modified CNN for classification. Model performance was evaluated with k -fold validation producing a strong performance with accuracy, sensitivity, and specificity $\geq 90\%$. While fusing models is an attractive deep learning architecture, the computational demands of combining ResNet-101 features with optimization-based CNN tuning raised questions for us about computing requirements and feasibility with larger datasets. We were also concerned by a lack of discussion of overfitting. Regularization strategies and loss-function analyses were also not thoroughly examined. For our project scope and target, it is important to understand the trade-off between performance and computational efficiency.

In 2025, Deng & Heath published work describing a feed-forward artificial neural network. [6] trained on the ImageNet 2012 contest dataset that achieved an ACC of 98.3% with a 99.69% Top-1 rate, and an average of 285.9 labels that are perfectly classified over the 10 batch partitions of the dataset.

The best performing model used 322,430,160 parameters with 4 decimal places precision. In 2023, the author also presented a new algorithm called gradient descent tunneling (GDT). The method uses stochastic gradient descent (SGD) to train a feedforward neural network (FNN) to a modest accuracy and then follows it up by training the model to zero error rate one data batch at a time, using the idea of homotopy from Numerical Analysis and Dynamical Systems—fields far from Machine Learning (ML). We find that, while the history of ImageNet-based outcomes is directly related to modern image-classifier architecture, this study uses a dataset that is small in size and does not address whether GDT can be effective for large datasets. Only an affirmative answer could preserve the utility of these findings for our project.

In 2026, a paper by Islam et al. [7] describes a deep learning-based tool created to address a lack of dermatology practitioners in the UK by providing faster and more accurate triage of skin cancer patients. The central methodology is the concatenation of enhanced metadata with pre-processed images, and the fusion of multiple EfficientNet-B2 models using all or part of the available images and metadata. The best model-fusion combination resulted in 99.50% sensitivity with 82.72% specificity and overall accuracy of 91.11%. Unfortunately, this study utilizes a non-public dataset, meaning peer-review of the study’s results will be minimal. Additionally, lack of diversity in the included patient files, especially among patients with darker skin tones may create bias and clinically employed models using this algorithm could underperform for ethnic populations less represented in the training data. It should be noted that while melanoma disproportionately affects lighter-skinned individuals, the rates of early detection for darker-skinned individuals lag and would benefit enormously from a more robust model design [4]. Finally, Islam et al. did not use the best available libraries for artefact removal and image preprocessing, resulting in visible reduction of image quality which the authors note.

Other articles we find helpful and are exploring:

- HAL [9]; EncoderDecodersTL [10]; LSNet [11]; MELdsPAPER [16] & MELds [15]
- mobilenetv2 [12]; NIH [13]; UNet [14] - augmentation strategy that may preserve fine-detail (lack possibly causing MEL miscat at 50%): "excessive augmentation" using "elastic deformations" of images. (p.3)
- AlexNet [17]: breakthrough paper that inspired ResNet, MobileNet, and EfficientNet architectures (RELU, dropout, GPU training, data augmentations, basic backbone: deep convolutional layers with maxpooling)
- Fitz [18]

IV. METHOD: ARCHITECTURE DESCRIPTION

Our approach applies a shared transfer learning strategy across three CNN models. Each backbone is initialized with ImageNet-pretrained weights and fully frozen to where only the classification head is trained. This head consists of a dropout regularization layer followed by a fully connected linear layer mapping to eight output classes. This structure is

consistent; the architectures initially differ only in backbone depth and the dimensionality of the learned feature vector passed to the head. Additionally, all models initially share identical training configurations before beginning ablation studies and HPO.

- 1) **ResNet-50 (RN)**: a 50-layer residual network using skip connections to mitigate vanishing gradients. Outputs a 2048-dimensional feature vector.
- 2) **MobileNetV3-Small (MN)**: a lightweight depthwise-separable architecture designed for resource-constrained deployment. Outputs a 1280-dimensional feature vector.
- 3) **EfficientNet-B0 (EN)** – a compound-scaled architecture balancing depth, width, and resolution. Outputs a 1536-dimensional feature vector.

A. Evaluation Metrics

Given the clinical context of skin lesion classification, we prioritize metrics that penalize false negatives. Missing a malignant lesion carries substantially greater risk than a false alarm. Our primary evaluation metrics are:

- **Accuracy (ACC)** – $(TP + TN)/n$
- **Loss** – $\hat{Y} - Y$
- **Balanced accuracy (BACC)** – mean per-class recall across all eight classes, weighted equally regardless of class size; our primary optimization target and the official ISIC 2019 challenge metric.
- **Macro F1** (future) – harmonic mean of precision and recall averaged equally across all classes; best single summary for imbalanced multiclass classification.
- **Per-class AUC**
- **Per-class sensitivity (TPR)** – (future) with particular attention to melanoma (MEL)
- **Per-class specificity (TNR)** – (future)

Confusion matrices will be generated for each model to visualize per-class performance. Note that ACC is reported for completeness only, as it is misleading under the severe class imbalance present in ISIC 2019. Model selection and fine-tuning is based primarily on macro F1 and BACC, which provide class-balanced performance estimates under severe label imbalance. In cases of comparable performance, melanoma (MEL) sensitivity is used as a secondary criterion to prioritize clinically critical error reduction.

B. Current Pipeline

- 1) **Data preparation.** Images and metadata are analyzed for class imbalance and diversity gaps. Minority class upsampled to address class imbalance. Split on `\texttt{lesion_id}` using `\texttt{StratifiedGroupKFold}` to ensure images from the same lesion do not appear in both training and validation sets, preventing data leakage.
- 2) **Image preprocessing.** Apply center crop to 224×224 (ImageNet V2 default), and normalize using ImageNet mean and standard deviation. Training augmentation includes random horizontal and vertical flips and color jitter (brightness, contrast, saturation) to improve generalization and improve edge detection.

- 3) **Model construction.** Pretrained weights loaded, backbone parameters are fully-frozen, and classification head (Dropout → Linear(9)) is attached. We have not incorporated an MLP for patient metadata at this time.
- 4) **Training.** Optimized with Adam and fixed learning rate (LR). LR scheduling is part of the ablation study.
- 5) **Evaluation.** Reported loss, ACC, BACC, macro F1, and per-class AUC-ROC, sensitivity, and specificity. Generated confusion matrices and per-class classification reports.
- 6) **Experiment tracking.** All runs logged to Weights & Biases (W&B) for metric comparison across model variants.

V. PRELIMINARY RESULTS

TABLE I: Preliminary results with minimal hyperparameter tuning. Shading indicates malignant condition (or potential). Each model was run for 10 epochs. Rivanna ran 4.5 hours for the 3 models and 10 epochs.

Metric	ResNet-50	MobileNetV3-Small	EfficientNet-B0
Loss	1.15	1.69	1.15
Val ACC	0.58	0.39	0.60
BACC	0.49	0.34	0.53
AUC	0.87	0.76	0.87
AUC: AK	0.88	0.83	0.89
AUC: BKL	0.76	0.61	0.79
AUC: BCC	0.90	0.78	0.89
AUC: DF	0.86	0.84	0.87
AUC: MEL	0.78	0.68	0.79
AUC: NV	0.88	0.81	0.89
AUC: SCC	0.90	0.75	0.88
AUC: VASC	0.97	0.81	0.95

TABLE II: Preliminary results with minimal hyperparameter tuning. Shading indicates malignant condition (or potential). Each model was run for 15 epochs on the full ISIC 2019 training dataset on a single GPU node (Rivanna HPC).

Metric	ResNet-50	MobileNetV2	EfficientNet-B3
Loss	1.24	1.32	1.31
Val ACC	0.62	0.56	0.54
BACC	0.56	0.53	0.52
Macro F1	0.46	0.39	0.38
AUC: AK	0.92	0.90	0.90
AUC: BKL	0.80	0.76	0.74
AUC: BCC	0.92	0.90	0.92
AUC: DF	0.89	0.87	0.87
AUC: MEL	0.82	0.81	0.81
AUC: NV	0.90	0.88	0.89
AUC: SCC	0.91	0.89	0.91
AUC: VASC	0.99	0.98	0.96

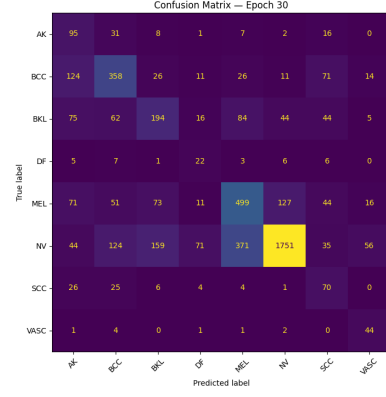


Fig. 1: Confusion matrix for ResNet50. Incorrect labeling of MEL is evident & most often confused with benign moles (NV).

We conducted initial transfer learning runs on a stratified 10% mini-split of the ISIC 2019 training data (local development) and full-dataset runs on the University of Virginia Rivanna HPC cluster using a single GPU node. Results confirmed expected behavior: all three models exhibit overfitting on the mini-split due to small dataset size and class imbalance, with training accuracy substantially exceeding validation accuracy. These runs validated our pipeline infrastructure and confirmed that class-weighted loss, dropout regularization, and early stopping are necessary baseline corrections before meaningful comparison is possible. Full baseline results on the complete dataset will be reported in the final submission.

VI. NEXT STEPS

A. Ablation Study of Three Base CNN Models

Using W&B for experiment tracking, we will systematically evaluate the contribution of individual design choices by removing or modifying one component at a time. Components under study include:

- **Class imbalance strategy** – we will compare three approaches: no correction (baseline), class-weighted loss (inverse-frequency weights applied to the cross-entropy loss function), and oversampling of minority classes via augmentation (random flips, rotation, and color jitter applied to underrepresented classes (DF: 0.9%, VASC: 1.0%, SCC: 2.5%, and AK: 3.4%) to bring their counts closer to the median class size).
- **Image pre-processing strategy** – fine tune a strategy for preprocessing of images which originate from the three sources. Our literature review revealed risks with center cropping-only as a strategy. Resizing and center cropping will therefore be compared to using black or average-value padding and resize to preserve aspect ratios.
- **Dropout rate** – varying the dropout probability before the classification head (0.0, 0.2, 0.3, 0.5).
- **Learning rate schedule** – step decay vs. cosine annealing vs. fixed learning rate.

- **Backbone freezing** – fully-frozen (FF) backbone (feature extraction only), partial extraction (PF) with fine tuning of last CNN layers, and full fine-tuning (UF) will be compared.

The key question for each component is whether its inclusion produces a measurable improvement in BACC and macro F1 on the validation set.

B. Hyperparameter Optimization (HPO)

Following ablation, we will apply Optuna for HPO with pruning, using W&B for tracking and Hydra for configuration management. Tunable parameters include learning rate, dropout probability, batch size, and augmentation intensity. Each of the three base architectures will be tuned independently to identify its optimal configuration prior to ensembling.

C. Ensembling

After fine-tuning three basemodels, we will evaluate all possible combinations (RN–EN, RN–MN, EN–MN, and RN–MN–EN).

D. Fusion Study

We hope to examine multimodal fusion (CNN + MLP) models wherein image and meta feature vectors would be concatenated prior to classification. Fusion variants would examine:

- 1) **Single CNN + MLP** – each base model paired individually with the metadata MLP branch.
- 2) **Dual CNN + MLP** – two backbone feature vectors concatenated with metadata features.

Using the same suite of evaluation metrics enables direct comparison with CNN-only models.

VII. MEMBER CONTRIBUTIONS

- **Haejin Son** – model development and testing, literature review, ablation study (focus: EfficientNet-B0)
- **Efrain Olivares** – model development and testing, method development, code storage pipelines and Git coordination, ablation study design. (focus: MobileNetV3-Small)
- **Bill Locke** – model development and testing, method development, ablation study and hyperparameter tuning research. (Focus: parallel model runs)
- **Virginia Brame** – model development and testing, literature review, fusion/ensembling model pipelines and image preprocessing strategies. (Focus: ResNet-50)

VIII. DECLARATION ON GENERATIVE AI

During the preparation of this work, the author(s) used Claude (Anthropic) and ChatGPT to: create a LaTeX draft using the provided template, help with setting up slurm workflow, and to provide help fusing and understanding pieces of code. The author(s) reviewed and edited all content as needed and take full responsibility.

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- [19] -----

- [20] *JUST TEMPORARY NOTES*: Thoughts about how not having enough fitzpatrick class 5 and 6 skin tones in the training data will affect results - despite image normalization. Complex issue since skin tone is not the only issue. Darker skinned individuals tend to have SC in places that are not exposed to the sun, so even the patterns of cause and effect can be different. Thoughts here (they will be incorporated into the final paper in a more organic fashion. This is just to keep them present and to connect them to certain other points in the paper.
- [21] JUST TEMPORARY NOTES*: Thoughts about the confusion of MEL and NV (also SCC and BCC) - see confusion matrix. Thoughts here (they will be incorporated into the final paper in a more organic fashion. This is just to keep them present and to connect them to certain other points in the paper.